

PharmaTwin

Digital Twin for Paracetamol Manufacturing

A proof-of-concept platform that watches a simulated tablet-manufacturing batch in real time, predicts where it's heading, and explains — in plain language — what to do about it.

1 plant modeled (Hyderabad)	12 process parameters tracked	5 batch KPIs predicted	8 working pages / modules
--	--	----------------------------------	--

CONTENTS

01 Project Overview & Objective	09 Historical Batch Comparison
02 Main Website Features	10 Root-Cause Analysis & Recommendations
03 Dashboard & KPIs	11 User Roles & Personas
04 Batch Explorer	12 Data & ML Flow
05 Process Monitoring	13 Technology Stack
06 KPI Prediction & Deviation Detection	14 Key Benefits & Use Cases
07 Alerts	15 Current POC Scope
08 AI Copilot	

01 Project Overview & Objective

PharmaTwin is a proof-of-concept "digital twin" for a Paracetamol 500mg tablet manufacturing line at the Hyderabad Plant. It is built around one reference batch

— the **Golden Batch**, a deliberately idealized, low-noise batch representing optimal process behavior — and uses it as the yardstick against which every other batch, live or historical, is measured.

The objective is to prove out, end to end, whether a plant can get an **early warning** when a batch is drifting away from Golden Batch performance, an automatically generated, plain-language **explanation of why**, and a **recommended corrective action** — before the batch finishes and the outcome is locked in. Because connecting to real plant sensors is a separate, larger undertaking, this POC proves the concept first against a realistic simulated production line, so the detection-to-recommendation loop can be validated safely before it is ever pointed at a live plant.

02 Main Website Features

Staff sign in with their Microsoft work account. Eight working areas make up the app today, but which of them you can navigate to depends on a persona switcher in the header (see [Section 11](#)) — no single persona sees all eight at once:

- **Dashboard** — role-specific summary of plant performance
- **Batch Explorer** — searchable record of every batch, finished or in progress
- **Golden Batch** — the reference profile everything else is measured against
- **Process Monitoring** — live, second-by-second view of a running batch
- **KPI Prediction & Deviation** — forecast of a running batch's likely final outcome
- **AI Review Desk** — inbox of every alert raised, with recommended actions to action
- **AI Copilot** — a chat assistant for asking plain-English questions about the floor
- **Settings** — start/stop simulated batches, choose fault scenarios for testing

HOW THE PERSONA SWITCHER GATES THESE

Plant Manager can reach Dashboard, Batch Explorer, Golden Batch, KPI Prediction & Deviation, AI Review Desk, AI Copilot, and Settings — not Process Monitoring. **Plant Operator** can reach Process Monitoring, Batch Explorer, AI Review Desk, AI Copilot, and Settings — not Dashboard, Golden Batch, or KPI Prediction. **Quality Engineer** can reach only Batch Explorer, AI Review Desk, AI Copilot, and Settings. Batch Explorer, AI Review Desk, AI Copilot, and Settings are the four pages common to all three.

03 Dashboard & KPIs

The Dashboard page is only ever reachable by the **Plant Manager** persona (see [Section 11](#)) — Plant Operator and Quality Engineer are automatically redirected elsewhere the moment that persona is selected, so in the live app there is exactly one Dashboard experience.

Plant Manager view

● Live data

Genuinely computed from the database, refreshed on load: **OEE %, Quality Score %, and Production (kg)**, each broken out by current shift / today / this month; a live count of active alerts; a chart of real OEE per completed batch against the Golden Batch's OEE; and running-batch / completed-today counts.

NOTE FOR PLANNING

The Manager dashboard's "Planned Runs" figure is currently a fixed placeholder number, not a real production schedule — there's no scheduling data source behind it yet.

04 Batch Explorer ● Live

A single, searchable table listing every batch — completed history and currently-running batches together — with search, status filtering, and sorting by yield or

quality.

- Opening a **completed** batch shows its full KPI breakdown (Yield, Output, Assay, Quality Score, Energy, OEE, Cycle Time), each with a plain-language "what does this mean" popup, plus a predicted-vs-actual comparison where a forecast exists.
- Opening a **running** batch shows a lighter live-status card and a link through to full detail in Process Monitoring.

DATA RETENTION

Completed batches for the current day accumulate up to 10 per plant. Once a day is no longer "today," it is automatically reduced to just the one real production batch recorded for that day — keeping the list meaningful long-term instead of filling up with demo activity.

05 Process Monitoring ● Live

The real-time control-room view of one running batch, refreshing every few seconds. All **12 tracked parameters** — temperature, pressure, flow rate, agitator speed, humidity, and seven others — are shown as cards, each comparing the current reading to the Golden Batch's expected band and flagging **Normal / Warning / Critical**.

Once a batch has run for 30 minutes, each parameter also gets a **30-minute-ahead forecast** from the trained machine-learning model, with its own predicted status. A headline banner always shows the single worst thing happening across the batch. Selecting any parameter opens a trend chart against the Golden Batch, a short history table, and an "Agent Assessment" — the automated root cause and recommended action for that parameter (see [Section 10](#)).

This page is reachable only by the **Plant Operator** persona — Plant Manager and Quality Engineer cannot navigate to it (see [Section 11](#)).

06 KPI Prediction & Deviation Detection ● Live

Separate from parameter-level forecasting, this predicts where the **whole batch** is likely to end up — its final Yield, Quality Score, Specific Energy Consumption, OEE, and Total Energy — using a machine-learning model trained on plant data, compared against the Golden Batch's actual final results.

- Each KPI is classified **Normal / Warning / Critical** against that benchmark.
- A confidence indicator is deliberately conservative early in a batch and firms up as more real data comes in.
- A "why" panel breaks the prediction down to the contributing process parameters and the underlying formula, not just a bare number — see [Section 10](#).

This page is reachable only by the **Plant Manager** persona — Plant Operator and Quality Engineer cannot navigate to it (see [Section 11](#)).

07 Alerts — the AI Review Desk ● Live

Every deviation the system detects — at the parameter level or the whole-batch KPI level — lands here automatically as one item in a single inbox, so nothing needs to be watched manually across separate screens.

Each alert carries a priority, a root cause, a recommended action, an urgency rating, a confidence score, and how long it's been sitting unreviewed. Staff can **Acknowledge** or **Reject** an alert. Alerts and their Acknowledge/Reject decisions are retained for **24 hours**, after which they are automatically purged regardless of which batch they belong to — there is no permanent alert archive today.

08 AI Copilot ● Live

A full-page chat assistant where anyone can type a plain-English question about what's happening on the floor — for one batch, several, or the whole plant — and get an answer.

- Answers are drawn only from real, current backend data (batch status, KPI predictions, parameter alerts, recommendations, historical comparisons) — the assistant is explicitly instructed never to invent a number or a cause.
- It always calls the underlying AI service live, regardless of the Static/AI-reasoning setting used elsewhere in the app (see [Section 10](#)).
- Answers now **stream in progressively**, word by word, instead of appearing all at once after a wait.
- **Remembers recent context** during a conversation, even if you return after being away for a while.

09 Historical Batch Comparison

• **Partially built** Every screen in the app already compares whatever you're looking at — a live batch, a completed batch, a single parameter — against the Golden Batch reference, and Batch Explorer shows a completed batch's forecast next to its actual outcome.

GAP

There is no dedicated screen yet for picking any two (or more) historical batches and viewing them side by side. Today, "comparison" always means "against the Golden Batch," not "against each other."

10 Root-Cause Analysis & Recommendations • Live

Every alert's explanation is produced the same way, whether it's about a single parameter or a whole-batch KPI:

1 Match against known fault patterns

The deviation is scored against a library of real historical fault signatures to rank the most likely cause.

2 Turn evidence into plain language

That evidence is written up as a root cause, recommended action, and business impact — either by a fixed template, or by asking the AI service to phrase the same evidence naturally.

3 Attach an honest confidence score

Every root cause and recommendation carries a High/Medium/Low confidence rating, based on how well the evidence actually matches historical precedent — never guessed.

The app can run in two reasoning modes, switchable in the UI:

MODE	WHAT GENERATES THE EXPLANATION
Static (current default)	A deterministic, rules-based write-up — fast, free, and consistent, no AI service call
Agent (AI)	The same evidence is handed to the AI service to phrase into a natural explanation, automatically falling back to the Static wording if the AI call fails

On the Process Monitoring and KPI Prediction pages, each explanation is visibly labeled which of the two actually produced it, so nothing is presented as more "AI-generated" than it really was.

11 User Roles & Personas

A switcher in the header lets anyone view the app as one of three personas — **Plant Manager, Plant Operator, or Quality Engineer** — and this controls which pages are reachable at all, not just what the Dashboard shows (see the page-access breakdown in [Section 02](#)).

IMPORTANT DISTINCTION

This is a **display/navigation preference, not a security boundary**. Sign-in today is Microsoft-account authentication only — there is no identity-based restriction, so anyone who successfully signs in can flip the persona switcher and reach every page and every action. The distinction is timing, not access: with a given persona currently selected, only that persona's pages are reachable; switching personas is what unlocks the rest. There is currently no admin/viewer distinction or permission system underneath it.

12 High-Level Data & ML Flow

The platform is built on a foundation of historical batch data, sensor-style time-series readings, and two purpose-trained machine-learning models:

A Historical foundation

120 recorded batches plus the purpose-built, idealized Golden Batch reference, each with minute-by-minute readings across 12 process parameters, feed everything downstream.

B Two trained forecasting models

One model predicts each parameter's value 30 minutes ahead; a second estimates a batch's likely final KPIs early, using the same trailing window of readings.

C Simulated live batches

A realistic minute-by-minute simulator stands in for a live plant feed for this POC, generating running batches that can be set to behave normally or to drift a chosen parameter for testing.

D Continuous evaluation

On every tick, fresh telemetry is checked against the Golden Batch envelope and the forecasting models; deviations raise alerts; the reasoning layer explains them.

E Recording the outcome

One genuine completed batch per plant per day is kept permanently as production history; the rest of the continuous demo activity is treated as disposable and cleared automatically.

13 Technology Stack

Backend	Python, FastAPI
Frontend	React, TypeScript, Tailwind CSS
Database	PostgreSQL
Machine learning	scikit-learn (Random Forest models)
AI orchestration	LangChain, coordinating tool calls and reasoning for the AI Copilot and the alert-explanation agents
AI reasoning & chat	Azure OpenAI, with Langfuse for monitoring AI calls
Sign-in	Microsoft Entra ID (Azure AD) single sign-on
Hosting	Single container on Azure Container Apps, via Azure Container Registry

Frontend and backend ship together as one lightweight container — a deliberately low-cost, easy-to-operate footprint appropriate for a proof of concept.

14 Key Benefits & Use Cases

- **Early warning, not hindsight** — flags individual process parameters up to 30 minutes before a threshold breach, and a batch's likely final quality/yield outcome from as early as 30 minutes into the run, instead of only finding out after the batch is already lost.
- **Explainable by default** — every alert comes with a plain-language cause, a recommended action, and an honest confidence score, so staff can prioritize instead of guessing.
- **Ask, don't hunt** — the AI Copilot answers plant-floor questions directly instead of requiring someone to dig through multiple dashboards, with answers appearing progressively rather than after a long wait.
- **One consistent yardstick** — every batch and parameter is judged against a real best-in-class reference batch, not an arbitrary spec sheet.

- **A safe proving ground** — because the whole detect-explain-recommend loop already runs end to end on realistic simulated data, it can be fully validated before it is ever connected to a live plant, de-risking that future step.

15 Current POC Scope

To keep expectations accurate, here is exactly where the platform stands today:

- **One plant, one product** — Hyderabad Plant, Paracetamol 500mg only. Not yet multi-plant or multi-product.
- **Simulated, not live, sensor data** — all "real-time" telemetry today comes from a realistic simulator, not a connection to real plant sensors, SCADA, or MES systems.
- **KPI-outcome model is directional, not certified** — it was trained on synthetic data and its accuracy is currently modest; it's used for comparison and demonstration, not as the system of record for a batch's real result.
- **Personas are cosmetic today** — no role-based access control yet; there's no identity-based restriction, so any signed-in user can switch personas and eventually reach everything, but which pages are reachable at a given moment is still gated by whichever persona is currently selected.
- **The Dashboard page is Plant Manager-only** — Plant Operator and Quality Engineer are automatically redirected away from it, so only the Plant Manager's live-data dashboard is ever seen in the live app.
- **State lives in memory** — user sessions and in-progress demo batches reset on a server restart; there's no persistent session store yet, which is fine at POC scale.
- **No side-by-side historical comparison screen** — comparison today always means "against the Golden Batch," not batch-to-batch.

EXECUTIVE SUMMARY

The short version

PharmaTwin is a working proof of concept that shows a plant floor can be watched in real time, given advance warning of a likely quality or yield problem — individual process parameters 30 minutes before a threshold breach, and a batch's likely final outcome from as early as 30 minutes into the run — and handed a plain-language, confidence-rated explanation and recommended fix, all without a human needing to piece it together from raw sensor data. Built for one plant and one product on realistic simulated production data, it already runs the full loop end to end: live monitoring, predictive KPIs, automatic alerting, AI-assisted root-cause explanation, and a conversational assistant staff can simply ask questions of — one that now answers progressively and keeps useful memory of a conversation even across a longer break.